Data Center Operational Blueprint

- User Workflow, User Experience, and Backend Implementation

Deployment Plan User Workflows (Background Information)

User workflow – Manual

BACKGROUND INFORMATION

1. View the cluster manifest
2. View whitespaces – select bundles and capacity cells
   - Review the rack capacity, cell power capacity, cooling capacity, redundancy, and cooling system for the capacity cell.
   - Consider the expansion options: will the tenant be able to expand to bundles
   - Create and start to build the placement plan: pick a capacity cell **or bundle** that may accommodate the cluster manifest (add/ remove CC/rack/bundle to a placement plan).
   - System performs some basic validation check if cc/bundle does not meet requirements. **System computes value and optionality cost**
3. Place Load
   - If fewer racks, manually enter; If more racks, download a template and upload csv.
   - Rack capacity check – load higher than rack capacity not allowed
   - **System computes value and optionality cost**
4. View resources and constraints
   - Use node data panel to see contained and upstream constraints
   - **Use dependency mapping on the spatial view to discover relationships.** Use topology to see relationships. **Focus on highlighting capacity and margins**
5. Adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - **Recompute/ update margins and costs**
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

BACKGROUND INFORMATION

The two levels of Placement/ Auto-placement

- There are two levels of decisions during load planning/ placement:
  1) **Macro-placement:** which capacity cell(s)/ bundles to use;
  2) **Micro-placement:** within the CC/bundle, how to distribute racks – which racks where
- Auto-placement also has these two layers.
  1) **Full auto-placement:** the algorithm determines both macro and micro placement
  2) **Partial auto-placement:** the user chooses the CC/bundle (macro-placement). The user may even specify the zones and rows. The algorithm places the racks optimally in the selected areas (micro-placement).

BACKGROUND INFORMATION

User workflow: partial auto-placement

1. View the cluster manifest
2. View whitespaces – select bundles and capacity cells
   - Review the rack capacity, cell power capacity, cooling capacity, redundancy, and cooling system for the capacity cell.
   - Consider the expansion options: will the tenant be able to expand to bundles
   - Create a placement plan
3. Auto-placement configuration
   - Launch auto-placement engine
   - Choose/ confirm capacity cells/ bundles**; Specify zone/row** if desirable. System performs some basic validation check if cc/bundle/zones do not meet requirements. **System computes value and optionality cost.**
   - Choose optimization rule
4. View results
   - Use node data panel to see contained and upstream constraints
   - Use dependency mapping on the spatial view to discover relationships. Use topology to see relationships. Focus on highlighting capacity and margins
   - **View value and optionality costs**
5. Manually adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - Recompute/ update margins and costs
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

BACKGROUND INFORMATION

User workflow: full auto-placement

1. View the cluster manifest
2. Auto-placement configuration
   - Launch auto-placement engine
   - Specify if any cc/ bundles/ racks shall NOT be used.
   - Choose optimization rule
3. **Auto-placement engine finds the optimal placement plan per configuration**
4. View results
   - Use node data panel to see contained and upstream constraints
   - **View value and optionality costs**
5. Manually adjust placement
   - Manually change kW numbers
   - Drag and drop rack loads – drag and drop rows and zones
   - Recompute/ update margins and costs
6. Save draft, save as another draft, save as plan.
7. Export csv of draft and plan

UX/UI

UX/UI Items

	Task Priority	Feature importance	Applicable to both Planning and Navigator?
Focus and dimming (not scene switching)	H	H	No
Persistent Labels	M	M	Yes
Expanding Labels to data badges	L	M	Yes
Actions from badges	L	M	Yes
Dependency tracer	L	H	Yes
Completion of data badge content	L	M	Yes, but different data
Load drag and drop	L	M	No

“Focus” (Frame Selection) and Dimming – Clarify Requirements

Effects of using Spatial Topology to navigate: **scene switching** (same as in Navigate: clicking on the nodes changes the scene.)

Effects of using Whitespace and Deployment Blueprints to navigate:

(A) **Focus – “Frame Selection”:** move the camera and adjust distance so the selected scope (as defined below in the table) fits nicely in view.

(B) **Selective dimming: dim the nodes not in the frame selection scope**

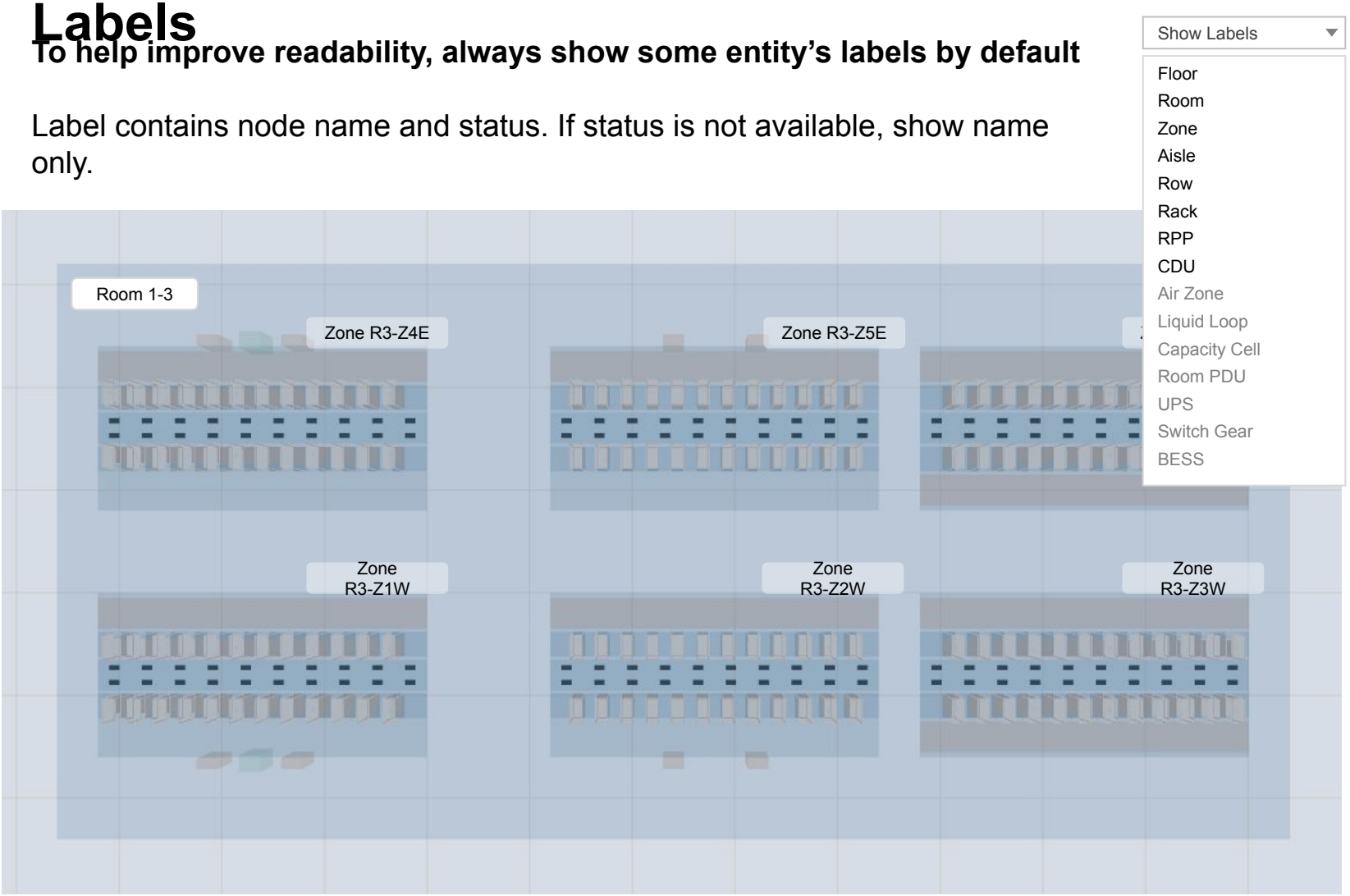
(C) **do NOT switching scenes, instead,** reset the scene to Site (meaning all the nodes should be in accessible if needed)

Selected Node Type in Whitespace and Deployment Navigations	**Frame Selection Scope**
Placement Plan	Entire site
Capacity Cell	All Rooms associated with the Capacity Cell and the substructure of these Rooms in the spatial topology, as well as SANs
Rack	The Row that contains the Rack
Containment Bundles	Entire site
Room Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs
Room PDU Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs
USP Bundle	All Rooms associated with the Bundle and the substructure of these Rooms in the spatial topology, as well as SANs

Persistent Labels

To help improve readability, always show some entity's labels by default

Label contains node name and status. If status is not available, show name only.



DEFAULT

(1) Show labels of
- Room
- Zone

(2) When these SANs are selected, also show their labels by default:
- Capacity Cell
- Air Zone
- Liquid Loop

(3) When a SAN is not selected, there is no option to show the label

Provide an option (drop down) for the user to show/hide labels by type.

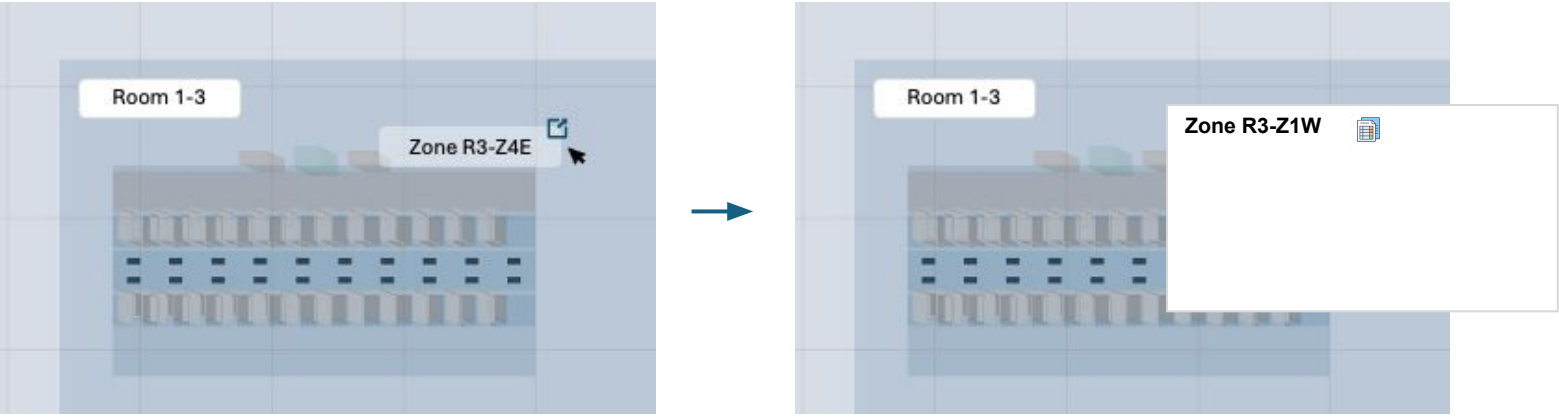
Labels not "shown" will still appear upon cursor over.

Expanding Labels to data badges

For each node, there are two types of annotated information: **labels and badges.**

- The label should contain only name and state. If state is not available, show name only.
- The badge should contain data. In the Planning view, show capacity and associated infrastructure data.
- **Default:** some node types by default show labels. Other node types by default show badges. Store this default setting as node data.

- When a label is visible, the user cursor over reveals the **expand icon**. Click on the icon to expand the label into a badge. This expansion is temporary to the interaction. The next time the node annotation behavior will go back to the default (i.e. a label).
- Clicking directly ON the label itself will open the node data panel.



Node Types with Default Data Badges in the Planning Section

- Rack
- Capacity Cell
- RPP
- Room PDU
- Air Zone
- Liquid Loop

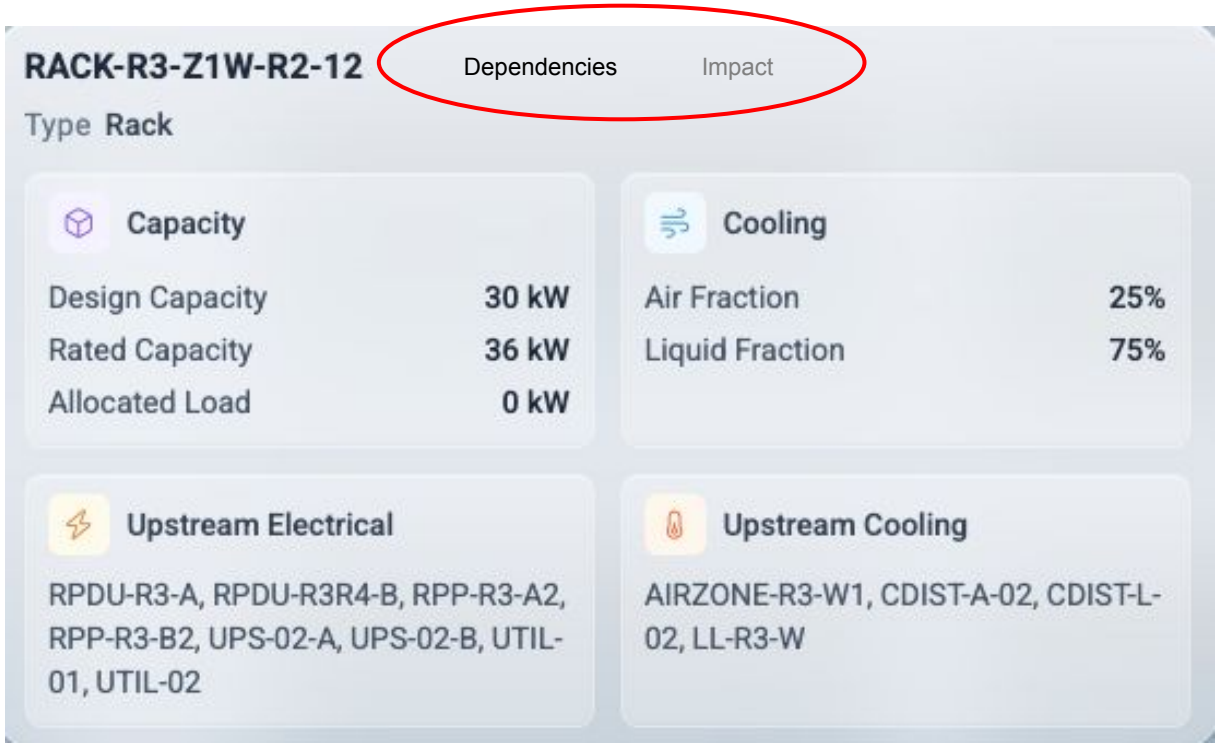
Other node types will show labels by default.

Currently we are not supporting an end-user option to change this default behavior, so the default will be determined at setup time as a configuration.

Actions from Badges

- Clicking on the badge itself opens the node data panel.

- Add two special actions: Trace Dependencies and Trace Impacts. These are functions of the Dependency Tracer



Dependency Tracer (WIP)

Dependency Tracer allows the users to highlight the **dependencies and impacts** of a selected node.

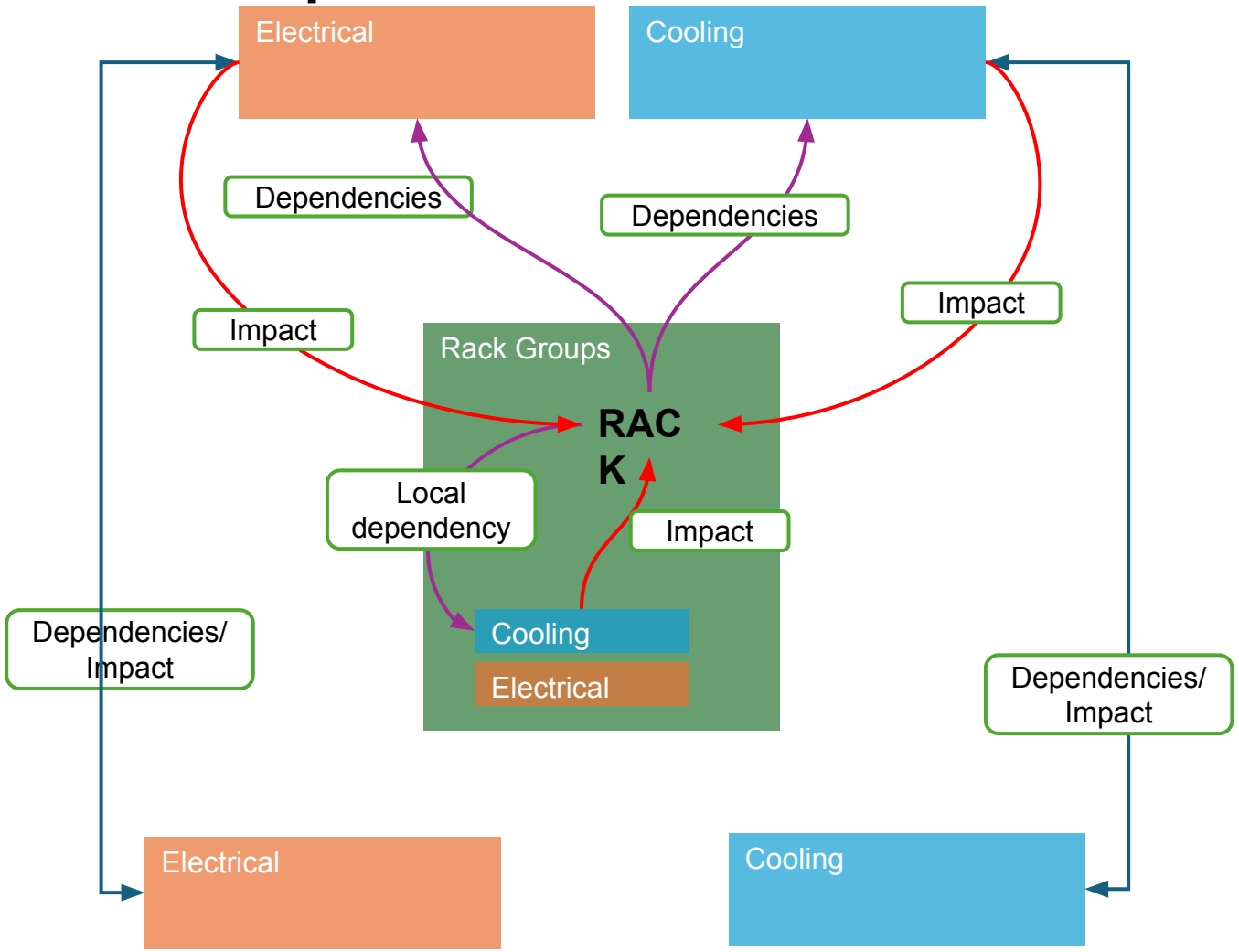
Example:

The user selected a rack, and clicked on "**Dependencies**".

- The Air zone and LL are highlighted (blue)
- The CDU and Air Cooling Units are also highlighted, if they are within view (blue)
- The RPP and Room PDU are highlighted (Red)

- **The trace should work in 3D view including the top view.**

Dependency Tracer Concepts



Rack Groups: entities that can be seen as a collection of racks

- Row
- Zone
- Capacity Cell
- Room
- Room Bundle
- Room PDU Bundle
- UPS Bundle

Impact and Dependency – two sides of the same coin

Impact

- Most nodes in the electrical and cooling systems can **impact** racks and rack groups Load Impact
- Most nodes in the electrical and cooling systems can also **impact** their own downstream nodes Downstream Impact

Dependency

- Rack and rack groups have **dependencies** on electrical and cooling system node. Some are Local Dependencies; some are Upstream Dependencies.
- Electrical and cooling nodes also have **dependencies** on their own upstream nodes Upstream Dependencies

Dependency Rules

For dependencies maps of the current topologies, see google sheet: https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=835908655#gid=835908655

Relevant Node Type	Dependency - Upstream	Dependency - Local
Rack	All electrical nodes upstream to the Rack PDUs of the Rack All cooling nodes upstream to the Rack	Rack PDU Cooling nodes downstream to the Rack
Row, Zone	Same as above	Same as above
Capacity Cell, Room Bundle	All electrical nodes upstream to RPP All cooling nodes upstream to Air Zone and LL	RPP Air Zone, LL
Room PDU Bundle, UPS Bundle	All electrical nodes upstream to UPS All cooling nodes upstream to Air Zone and LL	RPP, UPS Air Zone, LL
Electrical Nodes from Rack PDU and up	All electrical nodes upstream to the selected node	N/A
Cooling Nodes from RDHx/ DTC and up	All cooling nodes upstream to the selected node	N/A

Impact Rules

For impact maps of the current topologies, see google sheet: https://docs.google.com/spreadsheets/d/1-gKV23TcmoDeI9TCRfp6uii8TWyNC3tg9SH-E6Cnb0A/edit?gid=835908655#gid=835908655

Relevant Node Type	Impact - Load	Impact - Downstream
Electrical Nodes from Rack PDU and up which are Capacity Nodes	Rack Rack Groups	Electrical nodes downstream to the selected node
Electrical Nodes from Rack PDU and up which are NOT Capacity Nodes	N/A	Electrical nodes downstream to the selected node
Cooling Nodes from RDHx/ DTC and up which are Capacity Nodes	Rack Rack Groups	Cooling nodes downstream to the selected node
Cooling Nodes from RDHx/ DTC and up which are NOT Capacity Nodes	N/A	Cooling nodes downstream to the selected node

Dependency and Impact maps should be created as system data and remain updated if the topologies change

Capacity Nodes and Active Constraints

Capacity domains (or Capacity nodes) are nodes where you track *supply and load balance*. They represent a domain or boundary where you can say:
“X kW available, Y kW consumed.”

Some Capacity nodes are Active Constraints, meaning they themselves limit the capacity of others.
Other Capacity nodes are Passive Constraints, meaning they receive constraints being imposed by other nodes.

Node Type	Topology	Capacity Node (Capacity Domain)	ActiveConstraint
Rack PDU	Electrical System	FALSE	FALSE
RPP	Electrical System	TRUE	TRUE
Room PDU	Electrical System	TRUE	TRUE
UPS	Electrical System	TRUE	TRUE
BESS	Electrical System	TRUE	TRUE
Switch Gear	Electrical System	FALSE	FALSE
Utility Feed	Electrical System	TRUE	TRUE
Generator	Electrical System	TRUE	TRUE
RDHx	Cooling System	FALSE	FALSE
DTC	Cooling System	FALSE	FALSE
Air Zone	Cooling System	TRUE	TRUE
Liquid Lood	Cooling System	TRUE	TRUE
Air Cooling Unit	Cooling System	FALSE	FALSE
CDU	Cooling System	FALSE	FALSE
Cooling Distribution	Cooling System	TRUE	TRUE
Cooling Plant	Cooling System	TRUE	TRUE
Rack	Spatial Topology	TRUE	FALSE
Row	Spatial Topology	FALSE	FALSE
Zone	Spatial Topology	FALSE	FALSE
Aisle	Spatial Topology	FALSE	FALSE
Capacity Cell	Whitespace Blueprint	TRUE	FALSE
Room Bundle	Whitespace Blueprint	TRUE	FALSE
Room PDU Bundle	Whitespace Blueprint	TRUE	FALSE
UPS Bundle	Whitespace Blueprint	TRUE	FALSE

Capacity Nodes and Active Constraints may change if topologies change.

They will remain a manual configuration of the system data to be updated.

Passive constraints

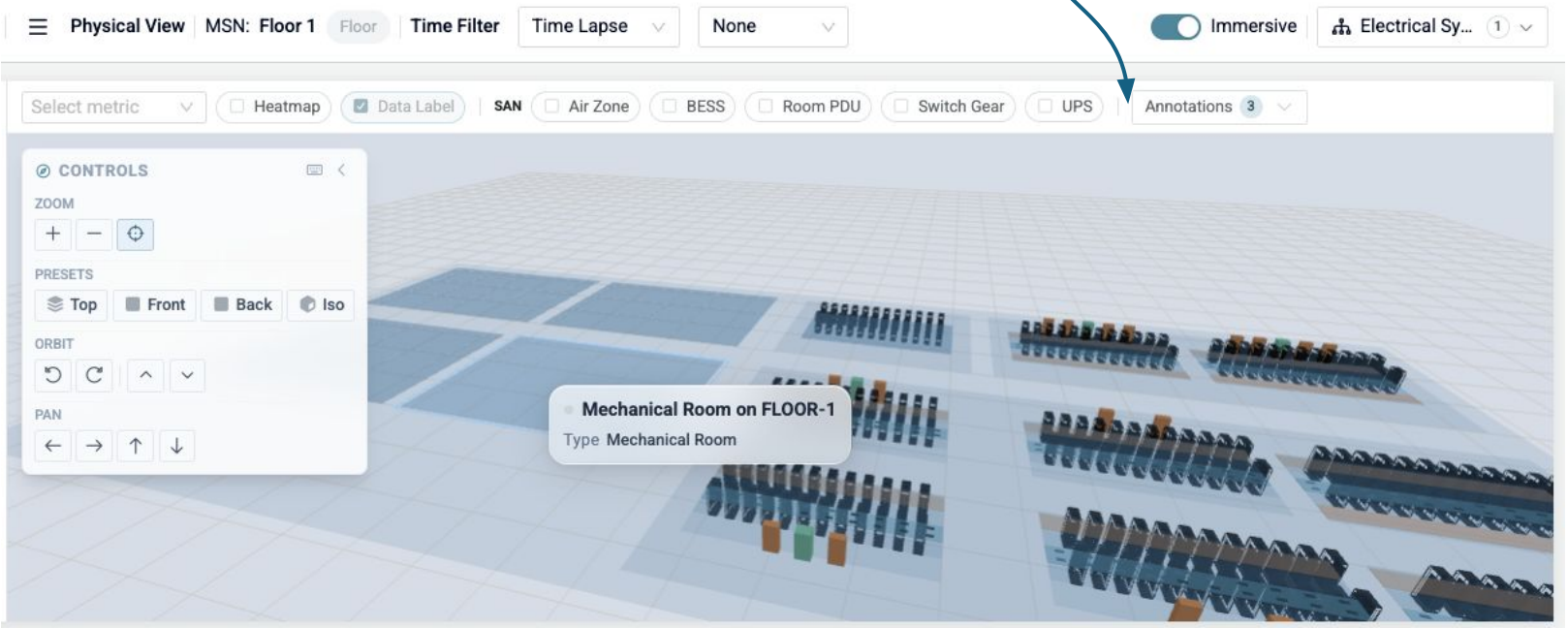
Two Ways to Show Dependencies/ Impacts

1. Interaction Bar

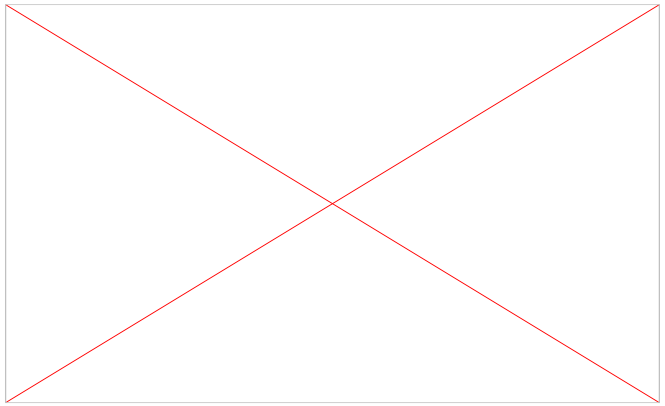
a. The user clicks and selects an entity in the 3D view (called “Source Node” or “Traced Source Node – TSN”) (the source node should be property highlighted)

b. The user clicks on the “Dependency” or ”Impact” button above.

Since the # of interaction buttons are growing, we may need to create a secondary layer, such as using tabs to show different menu items



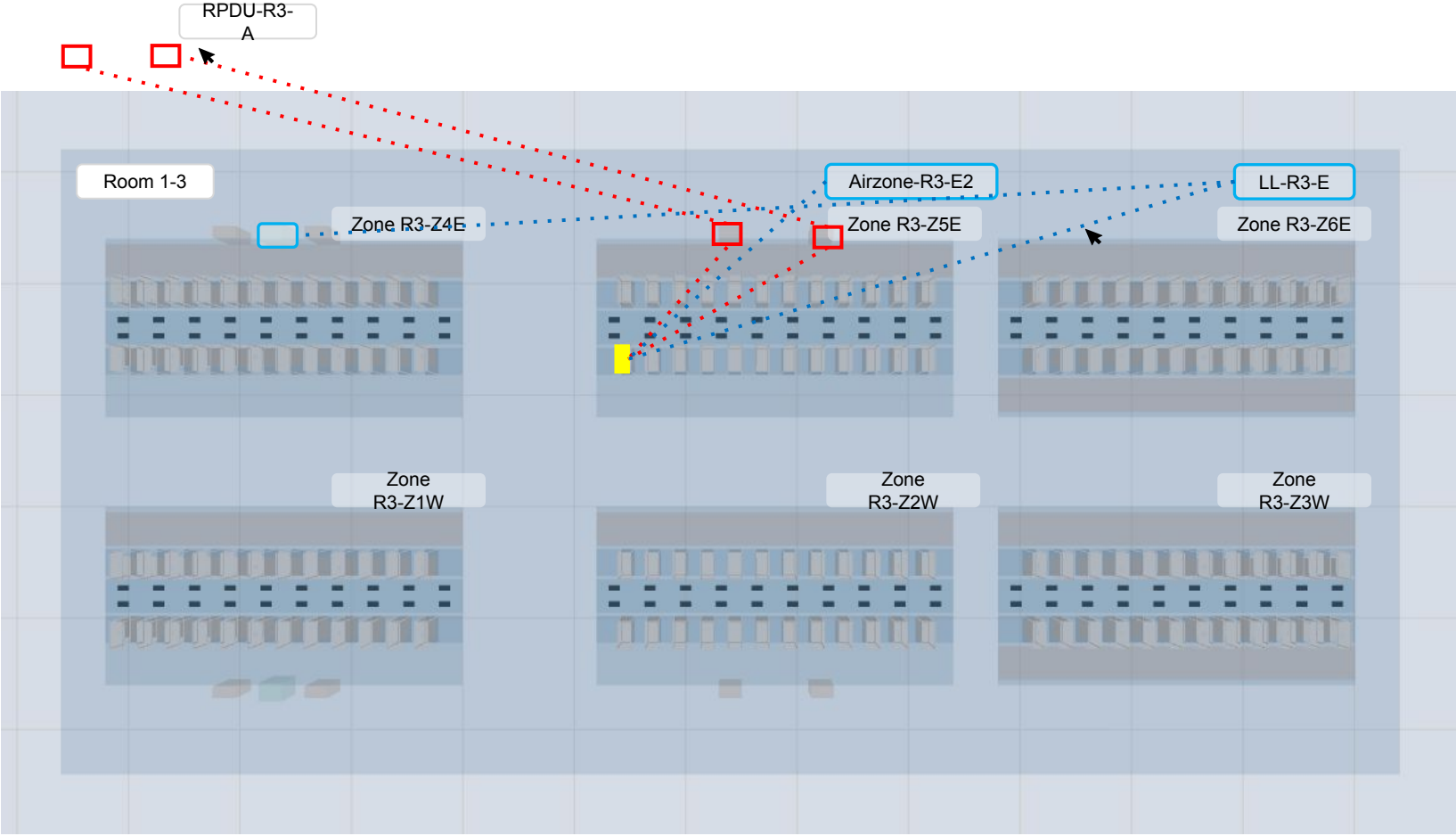
2. From the data badge of the DSN



Tracing Behaviors: Two Parts of of Dependency/ Impact Effects

Part 1 – contour highlight: the dependent or impacted entities have bright contour highlights

- Use different colors for different topology (e.g. red/orange for electrical, blue for cooling)
- For single geo/box entities, show contour around the object itself
- For SANs that are memberships (Air Zone, LL, Capacity Cell, the containment bundles), highlight both the label and the contour of the shape.
- For entities that are areas (row, zone, room), also highlight both the label and the contour of the shape.



Part 2 – tracing lines: lines that connect the entities according to topology (edges)

- Try to render tracing lines as native objects in the 3D view as well, so that they can be visible from different 3D angles.
- Straightlines are OK for now, no need to use curves/joints.
- For SANs that are memberships, and for entities that are areas, **connect the lines to the label, not to the shape**

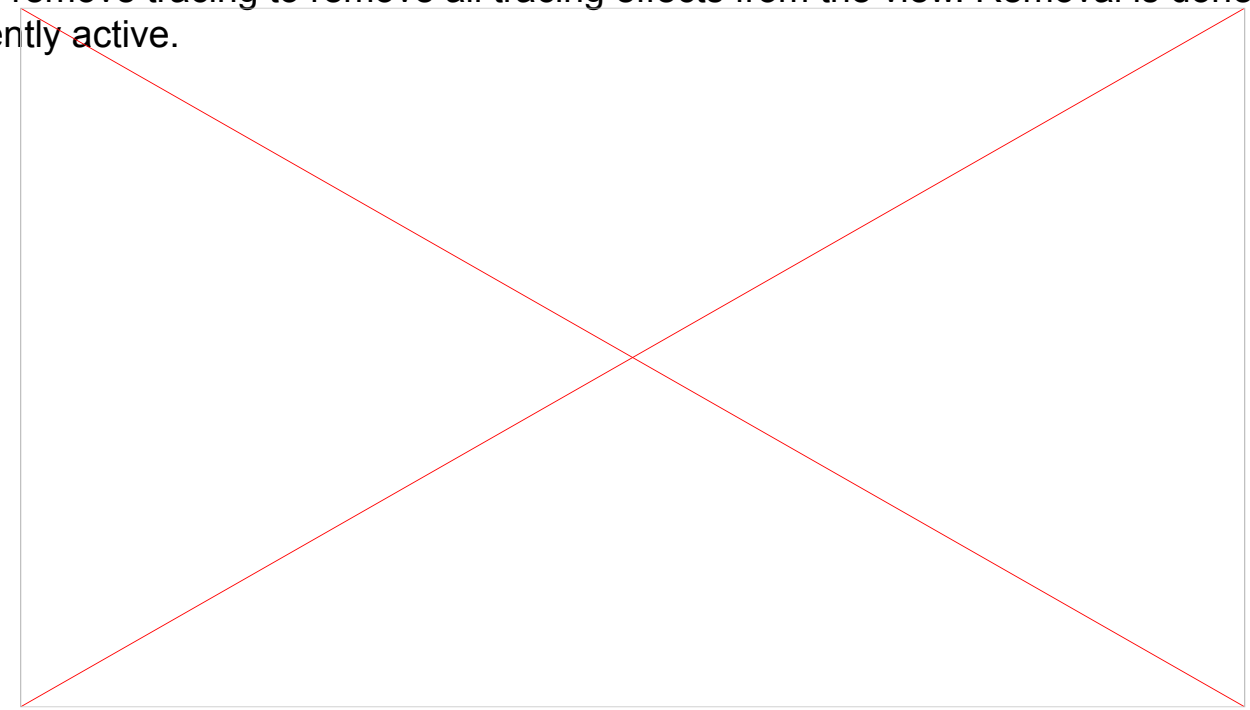
Tracing Behaviors

Tracing Levels

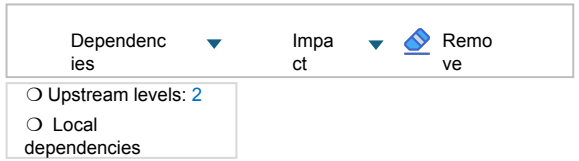
Upstream and downstream node/edges to the source node should carry a Level # in the data, which indicates how many degrees away they are from the Source Node. By default, **show two levels upstream and two levels downstream (**But this setting can be changed by the user)

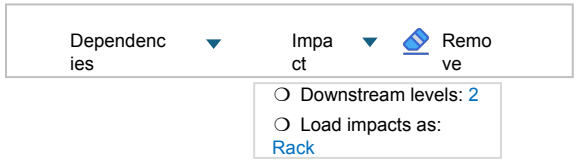
Multiple Tracing: Tracing can be shown for **multiple Source Nodes**. For example, once the upstream of a rack is traced, the user can continue on and select another rack to trace upstream, or they can select one of the upstream nodes (e.g. RPP), and rack its own upstream.

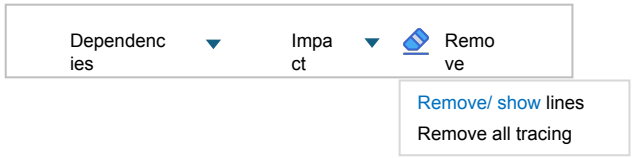
Remove Tracing: Click on remove tracing to remove all tracing effects from the view. Removal is done for dependencies/impacts of ALL source nodes that are currently active.



Tracing Options – these settings applies to **all active TSNs** at the same time







Dependency Options

- Any Traced Source Node (TSN) can have two types of dependencies: Upstream and Local. The user can choose to show both types or only one of them.
- If showing upstream dependencies, the user can also choose how many levels of upstream are shown.
- If local dependencies are checked, all local dependencies will be shown. There is no “level” with local dependencies.

Impact Options

- Any TSN can have two types of impacts: Downstream and Load Impacts. The user can choose to show both types or only one of them.
- If showing downstream impacts, the user can also choose how many levels of downstream are shown.
- “Load Impacts” means the TSN impacts racks and rack groups. If showing Load Impacts, the scope of the impacts has to be selected. Only one option can be chosen?
    - Rack: means the individual impacted racks are highlighted one by one
    - Row or Zone or Room: means to treat a row/ a zone/ a room as a single entity. If any rack within the row/zone/room is impacted, the entire group is considered impacted.
    - Capacity cell or room bundle, or room PDU bundle, or UPS bundle: means to treat the CC or the bundle as a single entity. If any rack within is impacted, the entire group is considered impacted.

- **Remove Options**
    - Remove all tracing means all highlights and lines will be removed
    - Remove lines means only the lines will be hidden, but the highlights will remain
    - Show lines meaning for the existing active TSN tracing only, bring back the lines

Load Drag and Drop

When the user edits rack load, enable drag-and-drop of the rack loads. After drop, provide options to either (1) reverse change or (2) update load (recompute)

Scope to be dragged:
Provide interaction so that the user can select a single rack, a single row, or an entire zone's rack loads at one time.

Places to drop:
A rack load can be dropped onto any rack with no load. A rack load cannot be dropped onto another rack with load.

- Validation: the load must be equal or smaller than the rack's circuit capacity

A row load can potentially be dropped onto another row, including partially loaded rows.

- Validation: the rack loads most be able to fit into the available empty racks in the target row.
  1) There has to be enough unloaded/free rack positions available
  2) The rack loads have to be able to fit into the available racks' circuit capacity. If these loads or capacity are not uniform across, then both of them have to be sorted from big to small. Then start by dropping the biggest load to the biggest capacity and work your way down.
  3) The spatial adjacency of the rack loads do not need to be maintained. Find available rack positions and fit in.

A zone can potentially be dropped onto another zone, including partially loaded zones.

- Validation: the rack loads most be able to fit into the available empty racks in the target zone.
  1) There has to be enough unloaded/free rack positions available
  2) The rack loads have to be able to fit into the available racks' circuit capacity. If these loads or capacity are not uniform across, then both of them have to be sorted from big to small. Then start by dropping the biggest load to the biggest capacity and work your way down.
  3) The spatial adjacency of the rack loads do not need to be maintained. Find available rack positions and fit in.

Backend (WIP)

Backend Items/ Modules

1. Load-Capacity Calculator
   - Uses the appropriate input data to compute capacity and load across nodes.
   - Some computations are (a) topology consolidation; Other computations involve (b) domain-specific logic
   - Create necessary triggers for the data updates
   - Validations for load placement
2. Variable data, data models
   - Clean up/ consolidate/ complete the variables
   - Other data models needed for backend
3. OP and Metrics
   - Clarify the technical difference between OP and Metrics and make changes if needed
4. Optionality value calculator
   - Compute optionality-adjusted kW value for any subdomain of the data center based on optionality rules.

Load Capacity Calculator

Backend Items/ Modules

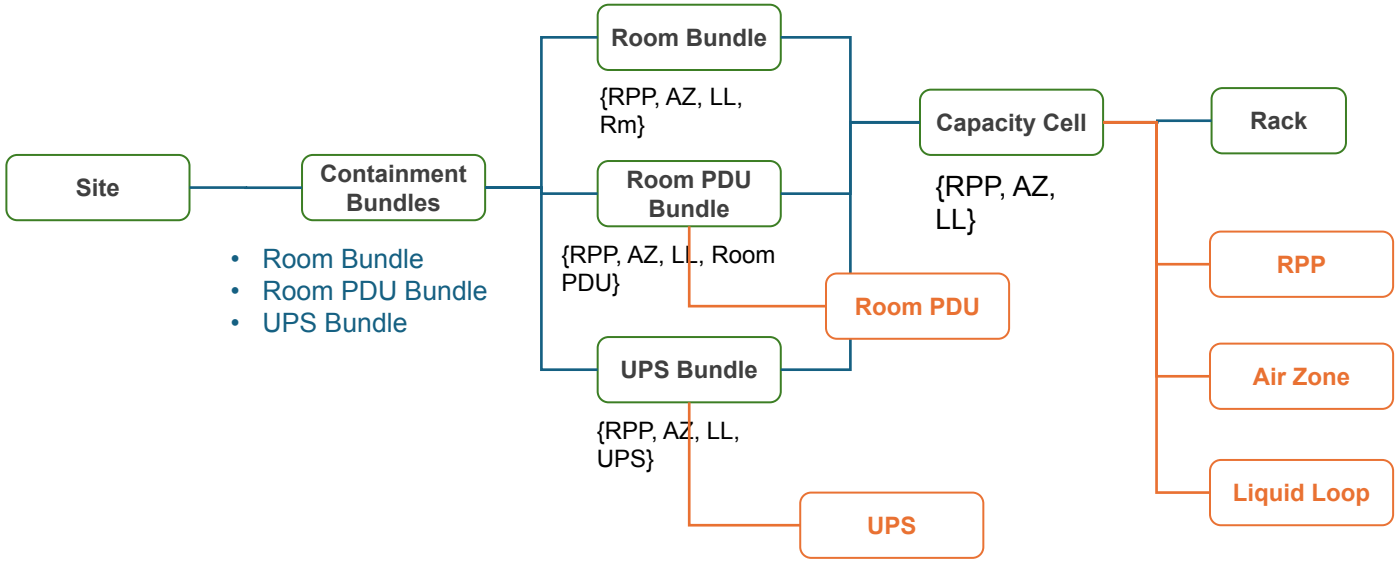
5. Auto-placement engine, module 1: Optionality Value Optimizer
6. Auto-placement engine, module 2: Capacity Packer
7. Blueprint Compiler
   - Given infrastructure input parameters, automatically organizes the data center into capacity cells and containment bundles
8. Topology update manager – if needed for future updates to be more scalable

	Requirements defined		
Load-Capacity Calculator	partial		
Variable data, data models			
OP and Metrics			
Optionality value calculator			
Auto-placement engine, module 1: Optionality Value Optimizer			
Auto-placement engine, module 2: Capacity Packer			
Blueprint Compiler			
Topology update manager			

Topology (WIP)

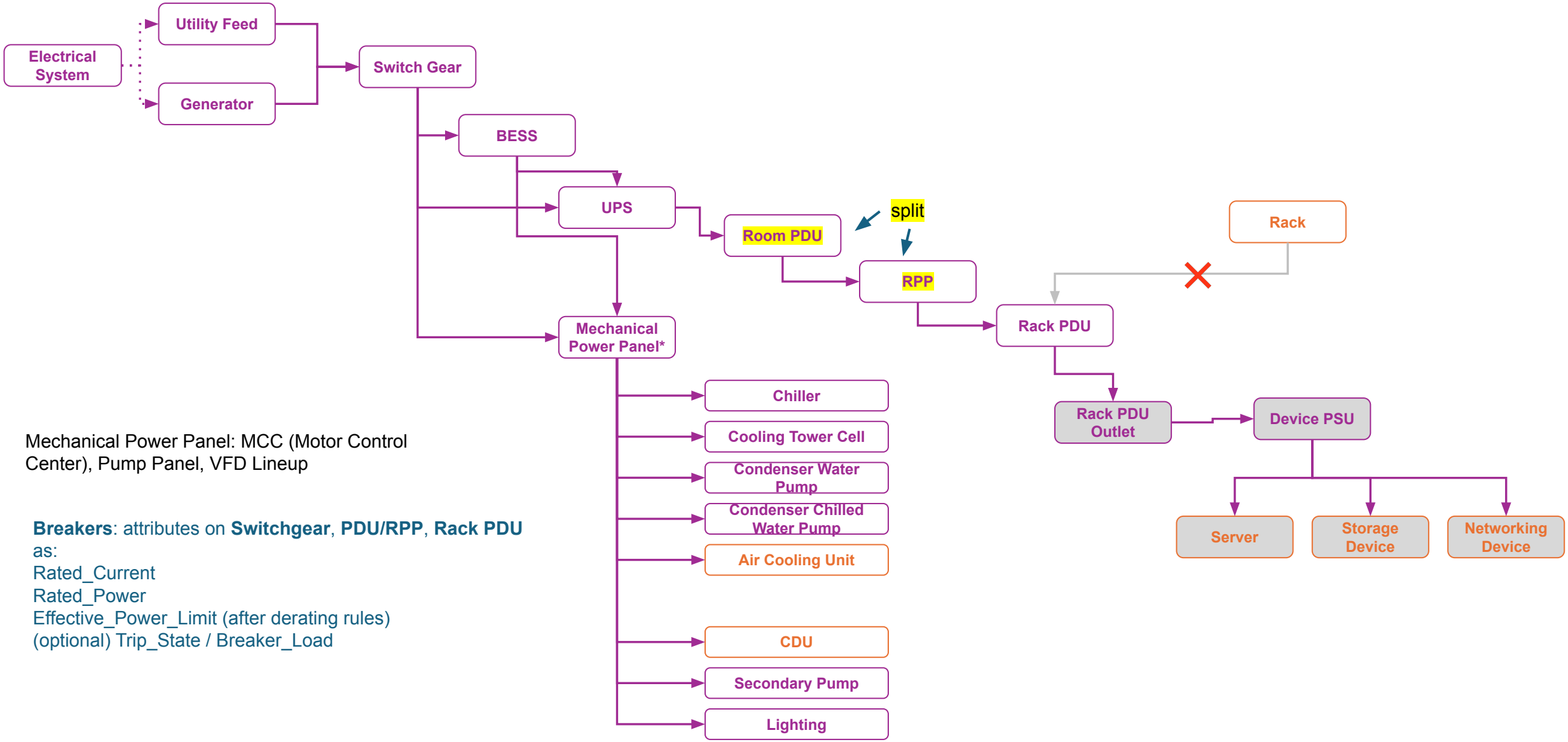
Whitespace Blueprint

The Whitespace will be ”carved” into capacity cells, then organized into containment bundles before the load placement planning starts



Electrical System (Multi-Context DAG)

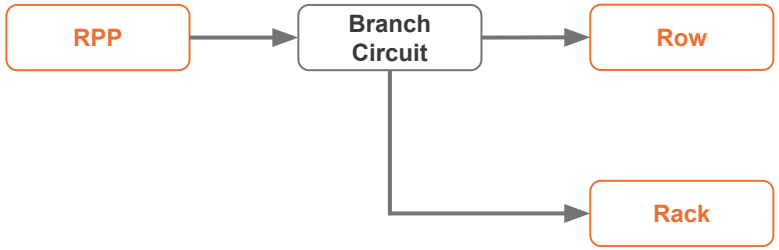
The gray node types may not be present in this topology during phase 3 planning.



Mechanical Power Panel: MCC (Motor Control Center), Pump Panel, VFD Lineup

Breakers: attributes on **Switchgear**, **PDU/RPP**, **Rack PDU** as:
Rated_Current
Rated_Power
Effective_Power_Limit (after derating rules)
(optional) Trip_State / Breaker_Load

Branch Circuit Topology



Branch Circuit modification

- The only first modification of the infrastructure we allow the user to do is to change the RPP to rack wiring.
- To better track it, we need to reflect the actual relationship by adding a Branch Circuit topology
  - RPP multiple branch circuits, each has a rating (e.g. 60A)
  - A number of branch circuits would be assigned to a row: Branch Circuit Row
  - Only at the load placement time, the branch circuits would be liked to the specific Rack: Branch Circuit Rack
  - In real deployment, branch circuit is wired to the Rack PDU. But for planning purposes, we can use Rack.
- During phase 3 (placement), we will have a default Branch Circuit Rack link. But will allow the user to change it if necessary.